Electronic Companion for the Paper

Event-Triggered Holomorphic Embedding Safe Reinforcement Learning for OPF in Integrated Electric-Gas Systems

Documentation for Integrated Electric-Gas System Topology Benchmarks

Version: June 2026

Prepared as supplementary documentation for the IEGS OPF reinforcement-learning benchmarks.

Contents

1 Data Sources

This document describes the data sources of the three integrated energy system subsystems used in the current experiments. The electric-network topologies are taken from MATPOWER and ACTIVSg power-system data[1]; the natural-gas network topologies are taken from the GasLib benchmark data[2].

2 Data Structure

2.1 Column Definitions for Gas-Network Data Tables

Table 1: Column Definitions of Gas-Network Data Tables.

Data table	Columns	Meaning and units
`Well`	0–6	gas-supply node, maximum/minimum gas production ($\mathrm{MSm^3/h}$), cost coefficient, gas-source index, reference-well flag, and gas production
`GenGas`	0–3	unit type, gas-consumption efficiency, gas node, and electric-side bus; type 1 denotes a gas turbine and type 0 denotes a conventional generation plant
`Demand`	0–23	24-period gas-load multipliers; the initial values are all ones and are replaced by the prescribed operating profile
`Load`	0–5	gas node, maximum pressure, minimum pressure, base gas load, node-type flag, and reference-node pressure
`Pipeline`	0–5	pipeline inlet node, pipeline outlet node, length (m), diameter (m), K_m-related parameter, and K_q-related parameter
`Comp`	0–14	inlet node, outlet node, type, compression factor, α, k_1, k_2, compression-ratio limits, electric-power limits (only for electric-driven compressors), electric-side bus, loss, efficiency, and current ratio
`P2G`	0–6	electric-side load bus, gas node, efficiency, current power (MW), base capacity (MVA), upper power limit (MW), and lower power limit (MW)

3 Benchmark Overview

Table 2: Sizes of the Three IEGS Benchmarks.

Case	Power buses count	Power loads count	Gen. units	GT units	Gas nodes count	Gas loads count	Pipes branches	Comp units	P2G units
`P14_G11`	14	11	5	2	11	3	8	2	1
`P57_G40`	57	42	7	2	40	29	39	6	1
`P200_G135`	200	108	49	6	135	99	141	29	9

[1] Zimmerman R D, Murillo-Sánchez C E, Thomas R J. MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education[J]. IEEE Transactions on Power Systems, 2010, 26(1): 12–19.

[2] Schmidt M, Aßmann D, Burlacu R, et al. GasLib–A library of gas network instances[J]. Data, 2017, 2(4): 40.

Table 3: Reinforcement-Learning State and Action Dimensions.

Case	State-space dimension	Action-space dimension
P14_G11	33	15
P57_G40	152	23
P200_G135	456	141

4 Calculation of Pipeline Coefficients K_m and K_q

4.1 Pipeline Model and Variables

For ordinary pipelines, the dynamic line-pack term and the steady-state flow term are characterized by coefficients K_m and K_q, respectively. For ordinary pipeline ℓ, let X_ℓ denote the pipeline length (m), D_ℓ the pipe diameter (m), $Z = 0.8$ the compressibility factor, $T = 281.15$ K the operating temperature, $G_s = 0.6106$ the relative density of natural gas, and $\epsilon = 0.05$ the pipe roughness parameter. Gas pressure is expressed in bar and gas flow is expressed in MSm^3/h.

To clarify the origin of these two coefficients, consider the simplified transient model of a horizontal gas transmission pipeline. After neglecting elevation and inertial terms, the continuity and momentum equations can be written as

$$\frac{\partial(\rho v)}{\partial x} + \frac{\partial \rho}{\partial t} = 0, \tag{1a}$$

$$\frac{\partial p}{\partial x} + \frac{f_\ell^{\mathrm{fr}} \rho v|v|}{2D_\ell} = 0, \tag{1b}$$

where ρ is the gas density, v is the in-pipe gas velocity, p is pressure, and f_ℓ^{fr} is the friction factor. Combining the gas equation of state with the standard-state volumetric-flow definition gives

$$p = \rho R T Z, \tag{2a}$$

$$\tilde{f}_\ell = \frac{\pi D_\ell^2}{4\rho_0} \rho v, \tag{2b}$$

where R is the gas constant, ρ_0 is the standard-state gas density, and $\tilde{f}_\ell$ is the standard-state volumetric flow. The pressure–flow form of the pipeline equations is then obtained as

$$\frac{\partial \tilde{f}_\ell}{\partial x} + \frac{\pi D_\ell^2}{4\rho_0 R T Z} \frac{\partial p}{\partial t} = 0, \tag{3a}$$

$$p\frac{\partial p}{\partial x} + \left(\frac{4}{\pi}\right)^2 \frac{f_\ell^{\mathrm{fr}} R T Z \rho_0^2}{2D_\ell^5} \tilde{f}_\ell |\tilde{f}_\ell| = 0. \tag{3b}$$

The factor $\pi/4$ originates from the circular cross-sectional area $A_\ell = \pi D_\ell^2/4$. In the flow equation, because velocity is expressed as $v \propto \tilde{f}_\ell / A_\ell$ and the friction term contains $v|v|$, the squared-area coefficient associated with $(\pi/4)^2$ appears.

4.2 Line-Pack and Weymouth-Flow Terms

Discretizing pipeline $\ell = (i, j)$ over its length X_ℓ, and representing the pipe state by the endpoint-pressure average $\bar{p}_{\ell,t} = (p_{i,t} + p_{j,t})/2$ and the average flow $\tilde{f}_{\ell,t} = (F_{\ell,t}^{\mathrm{in}} + F_{\ell,t}^{\mathrm{out}})/2$, yields the line-pack term and the

Weymouth-type flow term:

$$m_{\ell,t} = \Phi_\ell^m \frac{\pi X_\ell D_\ell^2}{4RTZ\rho_0}\bar{p}_{\ell,t}, \tag{4a}$$

$$m_{\ell,t} = m_{\ell,t-1} + F_{\ell,t}^{\text{in}} - F_{\ell,t}^{\text{out}}, \tag{4b}$$

$$\tilde{f}_{\ell,t}|\tilde{f}_{\ell,t}| = \Phi_\ell^q \left(\frac{\pi}{4}\right)^2 \frac{D_\ell^5}{X_\ell f_\ell^{\text{fr}} RTZ\rho_0^2}\left(p_{i,t}^2 - p_{j,t}^2\right). \tag{4c}$$

Here, Φ_ℓ^m and Φ_ℓ^q collect the conversions among pressure, time, and standard-volume units. If $\Sigma_{\ell,t} = F_{\ell,t}^{\text{in}} + F_{\ell,t}^{\text{out}}$ is introduced, then $\tilde{f}_{\ell,t} = \Sigma_{\ell,t}/2$. Therefore, the Weymouth relation in average-flow form can equivalently be written as $\Sigma_{\ell,t}|\Sigma_{\ell,t}| = 4K_{q,\ell}(p_{i,t}^2 - p_{j,t}^2)$.

4.3 Engineering-Unit Coefficients

Under the engineering units adopted in this document, the above line-pack and flow terms are folded into two tabulated coefficients, $K_{m,\ell}$ and $K_{q,\ell}$. The friction correction term is first computed as[3]

$$\Lambda_\ell = \left[2\log_{10}\left(\frac{3.7D_\ell}{\epsilon}\right)\right]^2, \tag{5}$$

where Λ_ℓ can be interpreted as an approximate expression for the reciprocal of the Darcy friction factor. After combining $1/f_\ell^{\text{fr}}$, standard-state parameters, and unit conversions, the pipeline coefficients are written as

$$K_{m,\ell} = C_m \frac{X_\ell D_\ell^2}{TZ}, \qquad K_{q,\ell} = C_q \frac{D_\ell^5 \Lambda_\ell}{ZTX_\ell G_s}. \tag{6}$$

The folded constants used in this document are

$$C_m = 2.119 \times 10^{-7}, \qquad C_q = 96.074830 \times 10^{-15}. \tag{7}$$

These two numerical constants result from combining standard-state parameters, pressure units, time units, and volume scaling. Taking the standard temperature as $T_0 = 273.15$ K, the standard pressure as $p_0 = 1.01325$ bar, and the air density as $\rho_{\text{air}} = 1.2922$ kg/m^3, the line-pack constant can be obtained from the ideal-gas equation of state and the circular pipe cross section:

$$C_m \simeq \frac{\pi}{4}\frac{T_0}{p_0} \times 10^{-9} = \frac{\pi}{4}\frac{273.15}{1.01325} \times 10^{-9} \approx 2.12 \times 10^{-7}, \tag{8}$$

where 10^{-9} accounts for the consistent conversion from standard volume to million standard cubic meters and for the length and area scales.

The flow-term constant C_q follows from the Weymouth-type pipeline-flow relation. When the squared-flow term is expressed in MSm^3/h and the squared-pressure term in bar^2, the part independent of pipe diameter, length, and friction can be written as

$$C_q = \left(\frac{\pi}{4}\right)^2 10^5 (3600)^2 \frac{T_0}{p_0 \rho_{\text{air}}} s_q, \tag{9}$$

where $(\pi/4)^2$ originates from the square of the circular cross-sectional area, 10^5 converts the pressure scale between bar and Pa, $(3600)^2$ converts squared flow from seconds to hours, $T_0/(p_0\rho_{\text{air}})$ converts between standard-state volume and mass density, and s_q collects the MSm^3 volume scale, the squared-pressure scale, and the length–diameter unit convention adopted by the gas-network data. Under the unit system used here,

$$s_q = 5.7606303 \times 10^{-28}, \tag{10}$$

therefore

$$C_q = \left(\frac{\pi}{4}\right)^2 10^5 (3600)^2 \frac{273.15}{1.01325 \times 1.2922} \times 5.7606303 \times 10^{-28} \approx 96.074830 \times 10^{-15}. \tag{11}$$

Thus, the tabulated K_m and K_q values are the pipeline coefficients obtained from the line-pack and Weymouth flow terms under the above unit system, given X_ℓ, D_ℓ, and Λ_ℓ.

[3] Humpola J. Gas network optimization by MINLP[M]. Logos Verlag Berlin GmbH, 2017; De Wolf D, Smeers Y. The gas transmission problem solved by an extension of the simplex algorithm[J]. Management Science, 2000, 46(11): 1454–1465.

4.4 Electric–Gas Energy Conversion

It should be distinguished that the energy–volume conversion coefficients for electric–gas coupling elements are based on a separate set of EIA energy-unit data[4], which is used to relate per-unit electric power to natural-gas volume. The simplified conversion adopted here is

$$1 \text{ MWh} \approx 3412.1416 \text{ ft}^3 \times 0.02831685 \approx 96.6 \text{ m}^3, \tag{12}$$

Therefore, when 100 MWh is used as the electric-power base,

$$\alpha_0 = \frac{96.6 \times 100}{10^6} = 9.66 \times 10^{-3} \text{ MSm}^3/(100 \text{ MWh}). \tag{13}$$

The electric–gas coupling coefficients of gas turbines, P2G units, and compressors are then obtained from α_0 by multiplying or dividing by the corresponding efficiency parameters.

5 `P14_G11`

5.1 Construction Summary

`P14_G11` consists of the IEEE 14-bus power network and an 11-node gas network. The power network is constructed using `pandapower.networks.case14()`; the 11-node gas network is a purpose-built small synthetic pipeline network for testing, containing three gas wells, three load nodes, eight ordinary pipelines, two compressors, and one P2G unit.

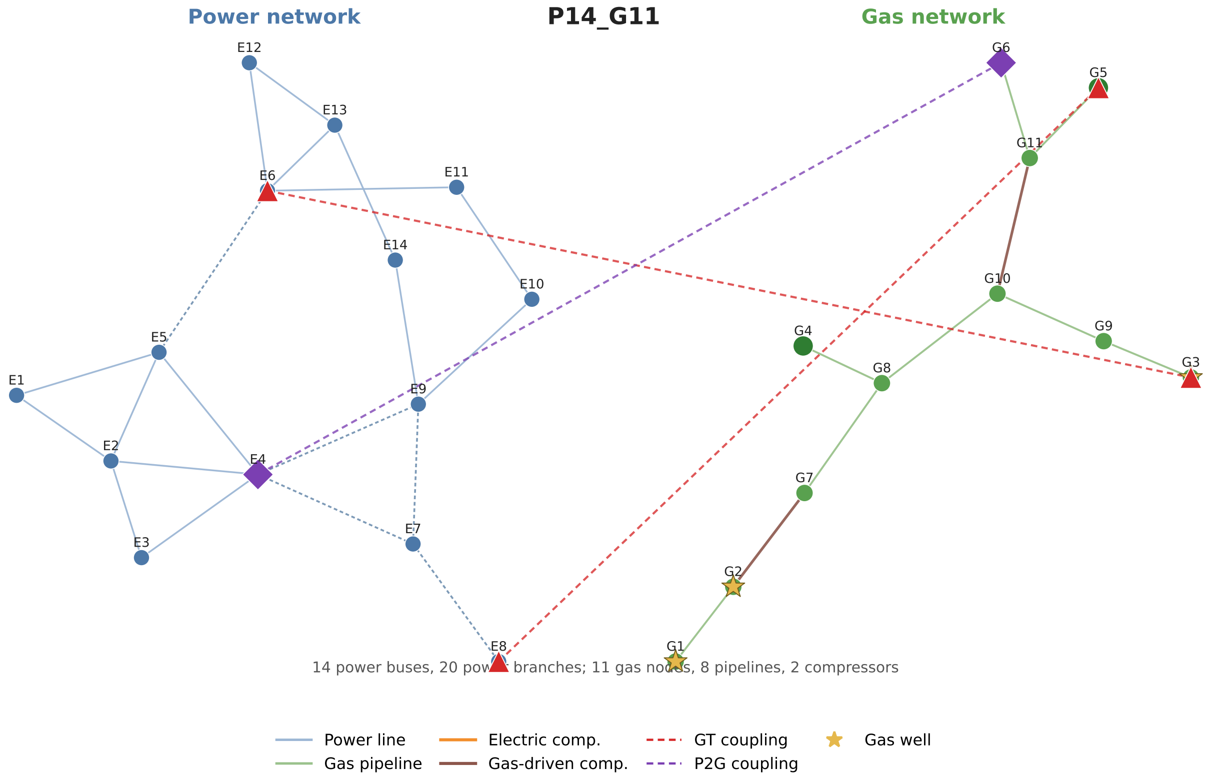

Figure 1: `P14_G11` Topology of the Integrated Electric–Gas System. In the figure, E denotes an electric bus and G denotes a gas-network node.

[4] EIA – Units & Calculators Explained, https://www.eia.gov/energyexplained/units-and-calculators/.

5.2 Gas-Side Data

This subsection reports the gas-side benchmark tables `Gas["Well"]`, `Gas["Load"]`, and `Gas["Pipeline"]`. The table `Gas["Load"]` corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

`Gas["Well"]`: Gas-Well Data

Table 4: `P14_G11` Gas-Well Data.

No.	$G_{\max}$ $\mathrm{MSm^3/h}$	$G_{\min}$ $\mathrm{MSm^3/h}$	Cost coeff.	Ref. well	Output $\mathrm{MSm^3/h}$
1	0.16	0	228.566	1	0
2	0.115	0	190.472	0	0
3	1	0	355.548	0	0

`Gas["Load"]`: Gas-Network Node Data

Table 5: `P14_G11` Gas-Network Node Data.

No.	Node	$p_{\max}$ bar	$p_{\min}$ bar	Base load $\mathrm{MSm^3}$ /h	Type	Pressure bar
1	1	70	40	0	1	50
2	2	70	40	0	3	0
3	3	70	40	0	3	0
4	4	70	40	0.1	2	0
5	5	60	40	0.12	2	0
6	6	60	40	0.08	2	0
7	7	70	40	0	0	0
8	8	70	40	0	0	0
9	9	70	40	0	0	0
10	10	70	40	0	0	0
11	11	70	40	0	0	0

`Gas["Pipeline"]`: Gas-network Pipeline Data

Table 6: `P14_G11` Gas-Network Pipeline Data.

No.	Pipeline inlet	Pipeline outlet	Length m	Diameter m	K_m $\mathrm{MSm^3}$ $\mathrm{h^{-1}}$ $\mathrm{Pa^{-1}}$	K_q $\mathrm{MSm^6}$ $\mathrm{h^{-2}}$ $\mathrm{bar^{-2}}$
1	1	2	55000	0.5	0.0152	2.09845×10^{-4}
2	7	8	55000	0.5	0.0188	4.74367×10^{-5}
3	3	9	55000	0.5	0.0129	6.89989×10^{-5}
4	8	4	55000	0.5	0.0118	7.58988×10^{-5}

No.	Pipeline inlet	Pipeline outlet	Length m	Diameter m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
5	8	10	55000	0.5	0.0271	1.18038×10^{-4}
6	9	10	55000	0.5	0.0129	6.89989×10^{-5}
7	11	5	55000	0.5	0.0188	4.74367×10^{-5}
8	11	6	55000	0.5	0.0188	4.74367×10^{-5}

5.3 Power-Side Units

This subsection lists all power-side generation units, including the reference-bus generator and the GT units coupled to the gas network.

Table 7: `P14_G11` Power-Side Generation Units.

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVAr	$Q_{\max}$ MVAr	V p.u.
1	1	GP	–	–	232.40	0.00	332.40	0.00	12.00	1.060
2	2	GP	–	–	40.00	14.00	140.00	-40.00	50.00	1.045
3	3	GP	–	–	0.00	14.00	100.00	0.00	40.00	1.010
4	6	GT	3	0.55	0.00	0.00	100.00	-6.00	24.00	1.070
5	8	GT	5	0.45	0.00	0.00	100.00	-6.00	24.00	1.090

5.4 Electric–Gas Coupling Elements

This subsection reports P2G units and compressors that couple the electric and gas subsystems.

Table 8: `P14_G11` P2G Units.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW
1	4	6	0.45	0.00	5.00

Table 9: `P14_G11` Compressor Units.

No.	Gas inlet	Gas outlet	Type	Power bus	Ratio p.u.	Elec. power range MW
1	2	7	Gas	6	1.00–1.20	–
2	10	11	Gas	6	1.00–1.20	–

6 P57_G40

6.1 Construction Summary

P57_G40 consists of the IEEE 57-bus power network and a 40-node gas network. This 40-node gas network roughly represents a segment of Germany's low-calorific gas transmission network serving the Rhine-Main-Ruhr region. The network contains three gas wells, 29 load nodes, 39 ordinary pipelines, six compressors, and one P2G unit.

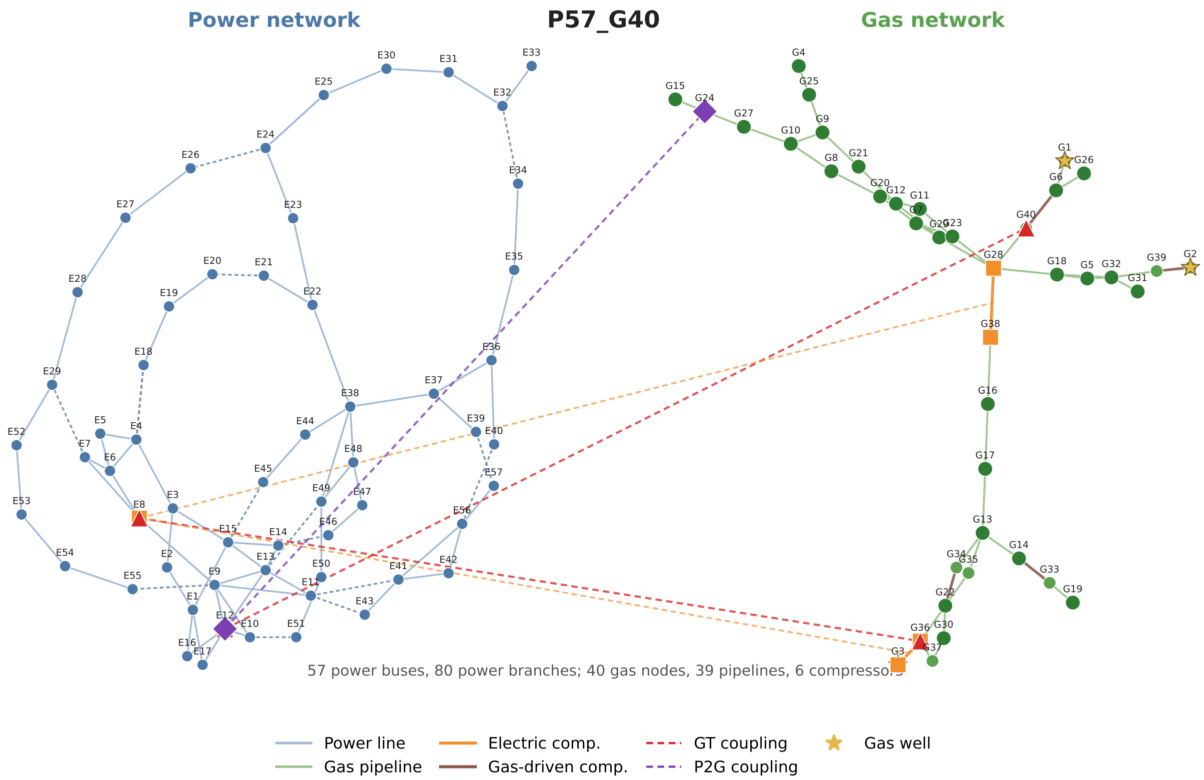

Figure 2: P57_G40 Topology of the Integrated Electric–Gas System. In the figure, E denotes an electric bus and G denotes a gas-network node.

6.2 Gas-Side Data

This subsection reports the gas-side benchmark tables Gas["Well"], Gas["Load"], and Gas["Pipeline"]. The table Gas["Load"] corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

Gas["Well"]: Gas-Well Data

Table 10: P57_G40 Gas-Well Data.

No.	$G_{\max}$ $\mathrm{MSm^3/h}$	$G_{\min}$ $\mathrm{MSm^3/h}$	Cost coeff.	Ref. well	Output $\mathrm{MSm^3/h}$
1	2.5	-1	2692.34	1	0
2	1.5	0	2243.61	0	0
3	1.5	0	2243.61	0	0

Gas["Load"]: Gas-Network Node Data

Table 11: P57_G40 Gas-Network Node Data.

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm^3 /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm^3 /h	Type	p_{ref} bar
1	81.0132	68	0	1	74	21	81.0132	30	0.075	2	0
2	81.0132	30	0	3	0	22	81.0132	30	0.075	2	0
3	81.0132	30	0	3	0	23	81.0132	30	0.075	2	0
4	81.0132	30	0.075	2	0	24	81.0132	30	0.075	2	0
5	81.0132	30	0.075	2	0	25	81.0132	30	0.075	2	0
6	81.0132	30	0.075	2	0	26	81.0132	30	0.075	2	0
7	81.0132	30	0.075	2	0	27	81.0132	30	0.075	2	0
8	81.0132	30	0.075	2	0	28	81.0132	30	0.075	2	0
9	81.0132	30	0.075	2	0	29	81.0132	30	0.075	2	0
10	81.0132	30	0.075	2	0	30	81.0132	30	0.075	2	0
11	81.0132	30	0.075	2	0	31	81.0132	30	0.075	2	0
12	81.0132	30	0.075	2	0	32	81.0132	30	0.075	2	0
13	81.0132	30	0.075	2	0	33	81.0132	30	0	0	0
14	81.0132	30	0.075	2	0	34	81.0132	30	0	0	0
15	81.0132	30	0.075	2	0	35	81.0132	30	0	0	0
16	81.0132	30	0.075	2	0	36	81.0132	30	0	0	0
17	81.0132	30	0.075	2	0	37	81.0132	30	0	0	0
18	81.0132	30	0.075	2	0	38	81.0132	30	0	0	0
19	81.0132	30	0.075	2	0	39	81.0132	30	0	0	0
20	81.0132	30	0.075	2	0	40	81.0132	30	0	0	0

Gas["Pipeline"]: Gas-network Pipeline Data

Table 12: P57_G40 Gas-Network Pipeline Data.

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
1	1	6	13071	1	0.0152	2.09845×10^{-4}	21	20	7	10586	0.6	0.0188	4.74367×10^{-5}
2	33	19	38447	0.8	0.0188	4.74367×10^{-5}	22	20	11	10452	0.6	0.0129	6.89989×10^{-5}
3	38	16	21558	1	0.0129	6.89989×10^{-5}	23	6	26	12397	0.8	0.0118	7.58988×10^{-5}
4	16	17	6998	1	0.0118	7.58988×10^{-5}	24	11	23	19303	0.6	0.0271	1.18038×10^{-4}

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
5	17	13	58219	0.8	0.0271	1.18038×10^{-4}	25	28	23	66037	0.6	0.0129	6.89989×10^{-5}
6	28	29	43345	0.8	0.0129	6.89989×10^{-5}	26	28	18	18969	1	0.0188	4.74367×10^{-5}
7	29	12	16579	0.6	0.0188	4.74367×10^{-5}	27	18	32	36061	0.8	0.0188	4.74367×10^{-5}
8	12	21	10023	0.6	0.0188	4.74367×10^{-5}	28	32	31	22224	0.8	0.0271	1.18038×10^{-4}
9	29	7	35219	0.6	0.0271	1.18038×10^{-4}	29	32	5	31180	0.8	0.0188	4.74367×10^{-5}
10	7	23	20322	0.6	0.0188	4.74367×10^{-5}	30	5	18	12767	1	0.0188	4.74367×10^{-5}
11	21	9	32868	0.8	0.0188	4.74367×10^{-5}	31	32	39	32921	0.8	0.0188	4.74367×10^{-5}
12	28	40	47488	0.8	0.0188	4.74367×10^{-5}	32	36	22	49866	0.8	0.0188	4.74367×10^{-5}
13	9	10	3803	0.6	0.0188	4.74367×10^{-5}	33	22	35	3479	0.8	0.0188	4.74367×10^{-5}
14	9	25	39036	1.8	0.0188	4.74367×10^{-5}	34	36	37	3418	1	0.0188	4.74367×10^{-5}
15	10	27	38660	1.8	0.0188	4.74367×10^{-5}	35	30	37	32449	1	0.0188	4.74367×10^{-5}
16	25	4	18018	1.8	0.0188	4.74367×10^{-5}	36	30	22	26427	0.8	0.0188	4.74367×10^{-5}
17	27	24	3068	1.8	0.0188	4.74367×10^{-5}	37	13	14	18137	1	0.0188	4.74367×10^{-5}
18	24	15	12016	1.8	0.0188	4.74367×10^{-5}	38	13	34	32528.5	0.8	0.0188	4.74367×10^{-5}
19	10	8	14043	0.4	0.0188	4.74367×10^{-5}	39	13	35	32766	0.8	0.0188	4.74367×10^{-5}
20	8	20	20635	0.6	0.0152	2.09845×10^{-4}							

6.3 Power-Side Units

This subsection lists all power-side generation units, including the reference-bus generator and the GT units coupled to the gas network.

Table 13: P57_G40 Power-Side Generation Units.

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVAr	$Q_{\max}$ MVAr	V p.u.
1	1	GP	–	–	128.90	0.00	575.88	-140.00	200.00	1.040
2	2	GP	–	–	0.00	0.00	100.00	-17.00	50.00	1.010
3	3	GP	–	–	40.00	0.00	140.00	-10.00	60.00	0.985
4	6	GP	–	–	0.00	0.00	100.00	-8.00	25.00	0.980
5	8	GT	36	0.55	450.00	0.00	550.00	-140.00	200.00	1.005
6	9	GP	–	–	0.00	0.00	100.00	-3.00	9.00	0.980
7	12	GT	40	0.45	310.00	0.00	410.00	-150.00	155.00	1.015

6.4 Electric–Gas Coupling Elements

This subsection reports P2G units and compressors that couple the electric and gas subsystems.

Table 14: P57_G40 P2G Units.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW
1	12	24	0.45	70.00	150.00

Table 15: P57_G40 Compressor Units.

No.	Gas inlet	Gas outlet	Type	Power bus	Ratio p.u.	Elec. power range MW
1	38	28	Electric	8	1.00–1.20	0–800
2	14	33	Gas	8	1.00–1.20	–
3	22	34	Gas	12	1.00–1.20	–
4	3	36	Electric	8	1.00–1.20	0–800
5	2	39	Gas	8	1.00–1.20	–
6	6	40	Gas	12	1.00–1.20	–

7 P200_G135

7.1 Construction Summary

P200_G135 consists of the ACTIVSg200 power network and a 135-node gas network. This 135-node gas network roughly represents a portion of Germany’ s high-calorific gas transmission network. The network contains six GT units, nine P2G units, and 29 compressors.

7.2 Gas-Side Data

This subsection reports the gas-side benchmark tables Gas["Well"], Gas["Load"], and Gas["Pipeline"]. The table Gas["Load"] corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference

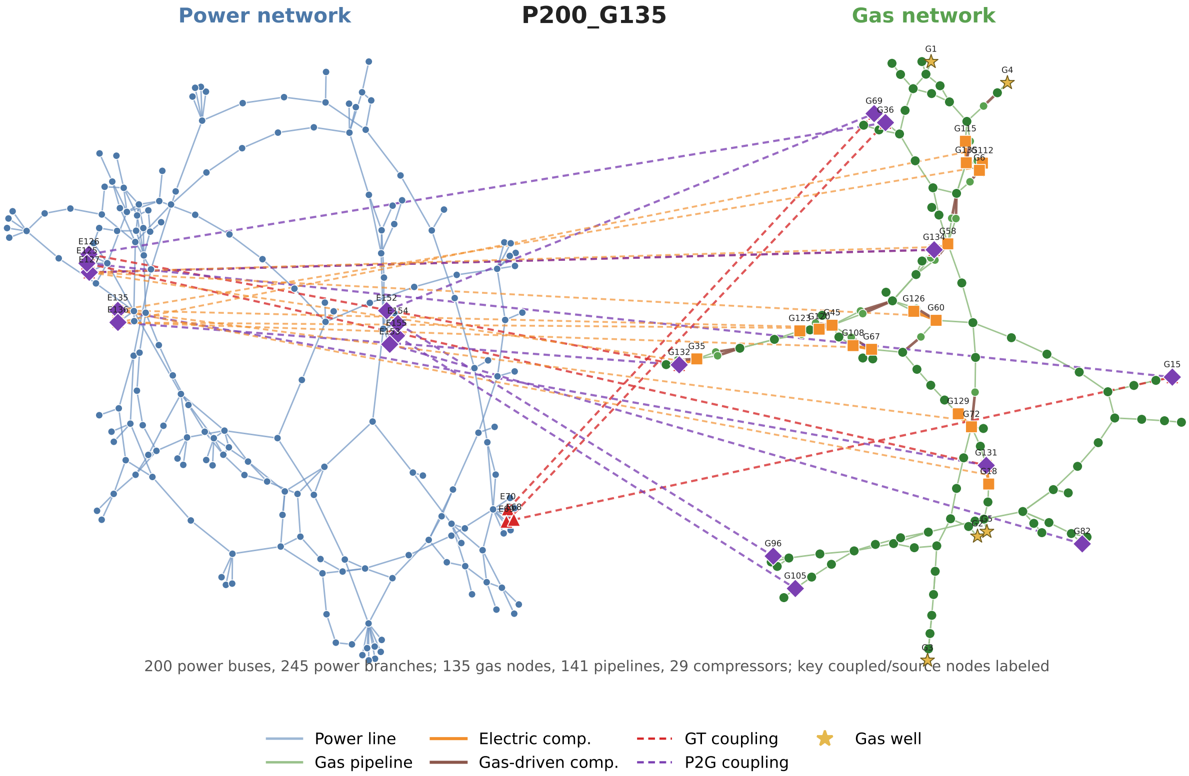

Figure 3: `P200_G135` Topology of the Integrated Electric–Gas System. In the figure, E denotes an electric bus and G denotes a gas-network node; only coupled nodes and gas-source nodes are labeled to avoid visual clutter.

node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

`Gas["Well"]`: Gas-Well Data

Table 16: `P200_G135` Gas-Well Data.

No.	$G_{\max}$ $\mathrm{MSm^3/h}$	$G_{\min}$ $\mathrm{MSm^3/h}$	Cost coeff.	Ref. well	Output $\mathrm{MSm^3/h}$
1	15	-15	4140	1	0
2	2	0	3450	0	0
3	2	0	3450	0	0
4	2	0	3450	0	0
5	2	0	3450	0	0
6	2	0	3450	0	0

`Gas["Load"]`: Gas-Network Node Data

Table 17: `P200_G135` Gas-Network Node Data.

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load $\mathrm{MSm^3}$ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load $\mathrm{MSm^3}$ /h	Type	p_{ref} bar
1	75	60	0	1	60	69	81.0132	30	0.04	2	0

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm^3/h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm^3/h	Type	p_{ref} bar
2	81.0132	30	0	3	0	70	81.0132	30	0.04	2	0
3	81.0132	30	0	3	0	71	81.0132	30	0.04	2	0
4	81.0132	30	0	3	0	72	81.0132	30	0.04	2	0
5	81.0132	30	0	3	0	73	81.0132	30	0.04	2	0
6	81.0132	30	0	3	0	74	81.0132	30	0.04	2	0
7	81.0132	30	0.04	2	0	75	81.0132	30	0.04	2	0
8	81.0132	30	0.04	2	0	76	81.0132	30	0.04	2	0
9	81.0132	30	0.04	2	0	77	81.0132	30	0.04	2	0
10	81.0132	30	0.04	2	0	78	81.0132	30	0.04	2	0
11	81.0132	30	0.04	2	0	79	81.0132	30	0.04	2	0
12	81.0132	30	0.04	2	0	80	81.0132	30	0.04	2	0
13	81.0132	30	0.04	2	0	81	81.0132	30	0.04	2	0
14	81.0132	30	0.04	2	0	82	81.0132	30	0.04	2	0
15	81.0132	30	0.04	2	0	83	81.0132	30	0.04	2	0
16	81.0132	30	0.04	2	0	84	81.0132	30	0.04	2	0
17	81.0132	30	0.04	2	0	85	81.0132	30	0.04	2	0
18	81.0132	30	0.04	2	0	86	81.0132	30	0.04	2	0
19	81.0132	30	0.04	2	0	87	81.0132	30	0.04	2	0
20	81.0132	30	0.04	2	0	88	81.0132	30	0.04	2	0
21	81.0132	30	0.04	2	0	89	81.0132	30	0.04	2	0
22	81.0132	30	0.04	2	0	90	81.0132	30	0.04	2	0
23	81.0132	30	0.04	2	0	91	81.0132	30	0.04	2	0
24	81.0132	30	0.04	2	0	92	81.0132	30	0.04	2	0
25	81.0132	30	0.04	2	0	93	81.0132	30	0.04	2	0
26	81.0132	30	0.04	2	0	94	81.0132	30	0.04	2	0
27	81.0132	30	0.04	2	0	95	81.0132	30	0.04	2	0
28	81.0132	30	0.04	2	0	96	81.0132	30	0.04	2	0
29	81.0132	30	0.04	2	0	97	81.0132	30	0.04	2	0
30	81.0132	30	0.04	2	0	98	81.0132	30	0.04	2	0
31	81.0132	30	0.04	2	0	99	81.0132	30	0.04	2	0
32	81.0132	30	0.04	2	0	100	81.0132	30	0.04	2	0
33	81.0132	30	0.04	2	0	101	81.0132	30	0.04	2	0
34	81.0132	30	0.04	2	0	102	81.0132	30	0.04	2	0
35	81.0132	30	0.04	2	0	103	81.0132	30	0.04	2	0
36	81.0132	30	0.04	2	0	104	81.0132	30	0.04	2	0
37	81.0132	30	0.04	2	0	105	81.0132	30	0.04	2	0
38	81.0132	30	0.04	2	0	106	81.0132	30	0	0	0
39	81.0132	30	0.04	2	0	107	81.0132	30	0	0	0
40	81.0132	30	0.04	2	0	108	81.0132	30	0	0	0

No.	p_{max} bar	p_{min} bar	Load MSm^3 /h	Type	p_{ref} bar	No.	p_{max} bar	p_{min} bar	Load MSm^3 /h	Type	p_{ref} bar
41	81.0132	30	0.04	2	0	109	81.0132	30	0	0	0
42	81.0132	30	0.04	2	0	110	81.0132	30	0	0	0
43	81.0132	30	0.04	2	0	111	81.0132	30	0	0	0
44	81.0132	30	0.04	2	0	112	81.0132	30	0	0	0
45	81.0132	30	0.04	2	0	113	81.0132	30	0	0	0
46	81.0132	30	0.04	2	0	114	81.0132	30	0	0	0
47	81.0132	30	0.04	2	0	115	81.0132	30	0	0	0
48	81.0132	30	0.04	2	0	116	81.0132	30	0	0	0
49	81.0132	30	0.04	2	0	117	81.0132	30	0	0	0
50	81.0132	30	0.04	2	0	118	81.0132	30	0	0	0
51	81.0132	30	0.04	2	0	119	81.0132	30	0	0	0
52	81.0132	30	0.04	2	0	120	81.0132	30	0	0	0
53	81.0132	30	0.04	2	0	121	81.0132	30	0	0	0
54	81.0132	30	0.04	2	0	122	81.0132	30	0	0	0
55	81.0132	30	0.04	2	0	123	81.0132	30	0	0	0
56	81.0132	30	0.04	2	0	124	81.0132	30	0	0	0
57	81.0132	30	0.04	2	0	125	81.0132	30	0	0	0
58	81.0132	30	0.04	2	0	126	81.0132	30	0	0	0
59	81.0132	30	0.04	2	0	127	81.0132	30	0	0	0
60	81.0132	30	0.04	2	0	128	81.0132	30	0	0	0
61	81.0132	30	0.04	2	0	129	81.0132	30	0	0	0
62	81.0132	30	0.04	2	0	130	81.0132	30	0	0	0
63	81.0132	30	0.04	2	0	131	81.0132	30	0	0	0
64	81.0132	30	0.04	2	0	132	81.0132	30	0	0	0
65	81.0132	30	0.04	2	0	133	81.0132	30	0	0	0
66	81.0132	30	0.04	2	0	134	81.0132	30	0	0	0
67	81.0132	30	0.04	2	0	135	81.0132	30	0	0	0
68	81.0132	30	0.04	2	0						

`Gas["Pipeline"]`: Gas-network Pipeline Data

Table 18: `P200_G135` Gas-Network Pipeline Data.

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
1	3	28	50304.8	1.2	0.0152	2.09845×10^{-4}	72	39	60	56432.5	1.2	0.0188	4.74367×10^{-5}

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
2	131	25	50170.3	1.2	0.0188	4.74367×10^{-5}	73	108	63	79066.7	1.2	0.0188	4.74367×10^{-5}
3	90	91	17722	1.2	0.0129	6.89989×10^{-5}	74	107	45	84957.9	1.2	0.0188	4.74367×10^{-5}
4	90	91	17722	1.2	0.0118	7.58988×10^{-5}	75	63	45	99697	1.2	0.0188	4.74367×10^{-5}
5	91	38	27576	1.2	0.0271	1.18038×10^{-4}	76	2	26	42281.5	1.2	0.0188	4.74367×10^{-5}
6	90	7	50039.1	1.2	0.0129	6.89989×10^{-5}	77	45	56	92797.1	1.2	0.0188	4.74367×10^{-5}
7	5	2	35005.3	1.2	0.0188	4.74367×10^{-5}	78	120	42	24981.3	1.2	0.0188	4.74367×10^{-5}
8	28	51	35640.1	1.2	0.0188	4.74367×10^{-5}	79	121	42	25743.4	1.2	0.0188	4.74367×10^{-5}
9	51	38	12536.2	1.2	0.0271	1.18038×10^{-4}	80	45	124	84458.2	1.2	0.0188	4.74367×10^{-5}
10	101	105	60939.7	1.2	0.0188	4.74367×10^{-5}	81	45	125	84946.7	1.2	0.0188	4.74367×10^{-5}
11	105	98	64597.2	1.2	0.0188	4.74367×10^{-5}	82	126	31	80621.8	1.2	0.0188	4.74367×10^{-5}
12	98	99	23354.6	1.2	0.0188	4.74367×10^{-5}	83	127	31	80904.3	1.2	0.0188	4.74367×10^{-5}
13	71	7	61293	1.2	0.0188	4.74367×10^{-5}	84	122	46	83083	1.2	0.0152	2.09845×10^{-4}
14	99	30	43259.3	1.2	0.0188	4.74367×10^{-5}	85	123	46	83957.7	1.2	0.0188	4.74367×10^{-5}
15	30	86	39290.8	1.2	0.0188	4.74367×10^{-5}	86	46	64	13264.2	1.2	0.0129	6.89989×10^{-5}
16	86	87	38503.4	1.2	0.0188	4.74367×10^{-5}	87	26	78	49223.1	1.2	0.0118	7.58988×10^{-5}
17	87	80	65028.4	1.2	0.0188	4.74367×10^{-5}	88	46	64	13264.2	1.2	0.0271	1.18038×10^{-4}
18	96	17	81181.6	1.2	0.0188	4.74367×10^{-5}	89	110	35	87864.1	1.2	0.0129	6.89989×10^{-5}
19	82	81	6510.82	1.2	0.0188	4.74367×10^{-5}	90	109	35	87454.8	1.2	0.0188	4.74367×10^{-5}
20	81	89	23219.5	1.2	0.0152	2.09845×10^{-4}	91	132	70	33438.2	1.2	0.0188	4.74367×10^{-5}
21	89	94	69525.7	1.2	0.0188	4.74367×10^{-5}	92	133	70	33473.9	1.2	0.0271	1.18038×10^{-4}

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
22	89	94	69525.7	1.2	0.0129	6.89989×10^{-5}	93	39	53	42519.4	1.2	0.0188	4.74367×10^{-5}
23	94	93	29067.2	1.2	0.0118	7.58988×10^{-5}	94	53	68	21281.5	1.2	0.0188	4.74367×10^{-5}
24	71	73	35220.9	1.2	0.0271	1.18038×10^{-4}	95	39	59	37246.5	1.2	0.0188	4.74367×10^{-5}
25	93	80	73550.2	1.2	0.0129	6.89989×10^{-5}	96	59	58	63409.2	1.2	0.0188	4.74367×10^{-5}
26	94	104	46081.6	1.2	0.0188	4.74367×10^{-5}	97	31	32	13796.9	1.2	0.0188	4.74367×10^{-5}
27	104	88	49928.8	1.2	0.0188	4.74367×10^{-5}	98	5	78	10397.6	1.2	0.0188	4.74367×10^{-5}
28	94	102	85858	1.2	0.0271	1.18038×10^{-4}	99	31	32	13796.9	1.2	0.0188	4.74367×10^{-5}
29	94	102	85858	1.2	0.0188	4.74367×10^{-5}	100	31	32	13796.9	1.2	0.0188	4.74367×10^{-5}
30	102	11	10634.5	1.2	0.0188	4.74367×10^{-5}	101	31	40	14097.1	1.2	0.0188	4.74367×10^{-5}
31	102	11	10634.5	1.2	0.0188	4.74367×10^{-5}	102	31	40	14097.1	1.2	0.0188	4.74367×10^{-5}
32	80	66	68792.3	1.2	0.0188	4.74367×10^{-5}	103	32	116	81461.5	1.2	0.0188	4.74367×10^{-5}
33	102	85	37563.1	1.2	0.0188	4.74367×10^{-5}	104	32	117	82408.8	1.2	0.0129	6.89989×10^{-5}
34	85	83	74204.5	1.2	0.0188	4.74367×10^{-5}	105	32	34	47648.2	1.2	0.0118	7.58988×10^{-5}
35	73	21	44307.6	1.2	0.0188	4.74367×10^{-5}	106	34	134	37524.1	1.2	0.0271	1.18038×10^{-4}
36	83	79	34864.5	1.2	0.0188	4.74367×10^{-5}	107	58	55	55637.1	1.2	0.0129	6.89989×10^{-5}
37	97	79	10819	1.2	0.0188	4.74367×10^{-5}	108	58	118	89446.2	1.2	0.0188	4.74367×10^{-5}
38	97	103	13882.6	1.2	0.0188	4.74367×10^{-5}	109	24	78	6844.75	1.2	0.0188	4.74367×10^{-5}
39	103	14	59093.6	1.2	0.0188	4.74367×10^{-5}	110	58	119	89179.2	1.2	0.0271	1.18038×10^{-4}
40	79	92	74213	1.2	0.0188	4.74367×10^{-5}	111	55	43	48033.2	1.2	0.0188	4.74367×10^{-5}
41	92	9	10380	1.2	0.0188	4.74367×10^{-5}	112	55	47	19465.6	1.2	0.0188	4.74367×10^{-5}

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
42	9	100	39573.1	1.2	0.0188	4.74367×10^{-5}	113	47	33	93300.9	1.2	0.0188	4.74367×10^{-5}
43	100	15	25443.5	1.2	0.0188	4.74367×10^{-5}	114	47	57	31776.6	1.2	0.0188	4.74367×10^{-5}
44	92	13	81583.8	1.2	0.0188	4.74367×10^{-5}	115	57	111	75196.4	1.2	0.0188	4.74367×10^{-5}
45	13	68	49535.2	1.2	0.0152	2.09845×10^{-4}	116	57	135	61179.7	1.2	0.0188	4.74367×10^{-5}
46	7	65	60573.5	1.2	0.0188	4.74367×10^{-5}	117	57	135	61179.7	1.2	0.0188	4.74367×10^{-5}
47	12	27	19740	1.2	0.0129	6.89989×10^{-5}	118	135	112	27318.1	1.2	0.0188	4.74367×10^{-5}
48	10	12	25479.1	1.2	0.0118	7.58988×10^{-5}	119	135	113	26466.9	1.2	0.0188	4.74367×10^{-5}
49	65	66	94839.2	1.2	0.0271	1.18038×10^{-4}	120	18	24	22659.1	1.2	0.0188	4.74367×10^{-5}
50	66	30	89395.5	1.2	0.0129	6.89989×10^{-5}	121	114	62	37046	1.2	0.0188	4.74367×10^{-5}
51	30	84	67770.9	1.2	0.0188	4.74367×10^{-5}	122	115	62	37887.8	1.2	0.0188	4.74367×10^{-5}
52	84	17	10478.4	1.2	0.0188	4.74367×10^{-5}	123	62	54	51079.8	1.2	0.0188	4.74367×10^{-5}
53	17	8	9209.88	1.2	0.0271	1.18038×10^{-4}	124	62	106	67517.7	1.2	0.0188	4.74367×10^{-5}
54	28	51	35640.1	1.2	0.0188	4.74367×10^{-5}	125	77	4	3000.01	1.2	0.0188	4.74367×10^{-5}
55	17	37	33603.3	1.2	0.0188	4.74367×10^{-5}	126	33	74	28202.2	1.2	0.0129	6.89989×10^{-5}
56	72	21	96065	1.2	0.0188	4.74367×10^{-5}	127	74	23	56443.9	1.2	0.0118	7.58988×10^{-5}
57	72	25	61286.2	1.2	0.0188	4.74367×10^{-5}	128	23	41	78489.8	1.2	0.0271	1.18038×10^{-4}
58	72	22	60740.5	1.2	0.0188	4.74367×10^{-5}	129	74	36	50533.9	1.2	0.0129	6.89989×10^{-5}
59	129	75	32476.9	1.2	0.0188	4.74367×10^{-5}	130	69	36	12358.4	1.2	0.0188	4.74367×10^{-5}
60	130	20	26313.7	1.2	0.0188	4.74367×10^{-5}	131	26	71	52638.3	1.2	0.0188	4.74367×10^{-5}
61	75	76	3274.84	1.2	0.0188	4.74367×10^{-5}	132	74	44	22865.5	1.2	0.0271	1.18038×10^{-4}

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
62	76	49	54491	1.2	0.0188	4.74367×10^{-5}	133	44	27	70219.5	1.2	0.0188	4.74367×10^{-5}
63	49	50	17417.5	1.2	0.0188	4.74367×10^{-5}	134	54	29	6799.79	1.2	0.0188	4.74367×10^{-5}
64	50	67	77016.1	1.2	0.0188	4.74367×10^{-5}	135	29	27	60895.6	1.2	0.0188	4.74367×10^{-5}
65	51	38	12536.2	1.2	0.0188	4.74367×10^{-5}	136	54	95	73579.3	1.2	0.0188	4.74367×10^{-5}
66	67	16	9158.22	1.2	0.0188	4.74367×10^{-5}	137	95	61	7053.13	1.2	0.0188	4.74367×10^{-5}
67	67	16	9158.22	1.2	0.0271	1.18038×10^{-4}	138	27	61	44719	1.2	0.0188	4.74367×10^{-5}
68	67	52	8869.54	1.2	0.0188	4.74367×10^{-5}	139	61	19	38054.1	1.2	0.0188	4.74367×10^{-5}
69	67	52	8869.54	1.2	0.0188	4.74367×10^{-5}	140	61	1	15990.8	1.2	0.0188	4.74367×10^{-5}
70	20	39	86829.9	1.2	0.0188	4.74367×10^{-5}	141	81	48	9159.01	1.2	0.0188	4.74367×10^{-5}
71	128	60	85824.9	1.2	0.0188	4.74367×10^{-5}							

7.3 Power-Side Units

This subsection lists all power-side generation units, including the reference-bus generator and the GT units coupled to the gas network.

Table 19: `P200_G135` Power-Side Generation Units.

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVAr	$Q_{\max}$ MVAr	V p.u.
1	49	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
2	50	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
3	51	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
4	52	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
5	53	GP	–	–	2.72	2.72	7.80	-1.11	4.23	1.040
6	65	GP	–	–	86.50	86.50	86.50	-21.66	32.04	1.040
7	67	GP	–	–	1.41	1.41	4.04	-0.57	2.19	1.040
8	68	GT	15	0.48	8.38	8.38	27.92	-3.41	13.01	1.040
9	69	GT	36	0.46	8.38	8.38	27.92	-3.41	13.01	1.040
10	70	GT	69	0.45	8.38	8.38	27.92	-3.41	13.01	1.040
11	71	GP	–	–	8.38	8.38	24.01	-3.41	13.01	1.040

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVAr	$Q_{\max}$ MVAr	V p.u.
12	72	GP	–	–	8.38	8.38	24.01	-3.41	13.01	1.040
13	73	GP	–	–	8.38	8.38	24.01	-3.41	13.01	1.040
14	76	GP	–	–	1.20	1.20	3.44	-0.44	2.04	1.040
15	77	GP	–	–	0.72	0.72	2.06	-0.27	1.22	1.040
16	78	GP	–	–	0.00	5.40	18.00	-2.00	9.16	1.040
17	79	GP	–	–	0.00	5.40	18.00	-2.00	9.16	1.040
18	90	GP	–	–	0.96	0.96	2.75	-0.36	1.63	1.040
19	91	GP	–	–	1.50	1.50	4.30	-0.55	2.55	1.040
20	92	GP	–	–	0.00	1.89	6.30	-0.70	3.21	1.040
21	94	GP	–	–	5.40	5.40	15.48	-2.20	8.39	1.040
22	104	GP	–	–	67.60	67.60	67.60	-14.26	21.09	1.040
23	105	GP	–	–	154.80	154.80	154.80	-28.51	42.17	1.040
24	114	GP	–	–	1.40	1.40	1.40	-0.24	0.36	1.040
25	115	GP	–	–	133.50	133.50	133.50	-21.60	31.95	1.040
26	125	GT	131	0.55	39.02	39.02	130.05	-15.87	60.60	1.040
27	126	GT	132	0.52	39.02	39.02	130.05	-15.87	60.60	1.040
28	127	GT	134	0.50	39.02	39.02	130.05	-15.87	60.60	1.040
29	135	GP	–	–	133.92	133.92	383.90	-54.46	208.02	1.040
30	136	GP	–	–	133.92	133.92	383.90	-54.46	208.02	1.040
31	147	GP	–	–	92.40	92.40	92.40	-14.47	21.41	1.040
32	151	GP	–	–	1.62	1.62	4.64	-0.66	2.52	1.040
33	152	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
34	153	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
35	154	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
36	155	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
37	161	GP	–	–	0.00	41.58	138.60	-15.38	70.55	1.040
38	164	GP	–	–	0.00	3.60	12.00	-1.33	6.11	1.040
39	165	GP	–	–	0.00	7.80	26.00	-2.89	13.23	1.040
40	166	GP	–	–	0.00	2.82	9.40	-1.04	4.78	1.040
41	167	GP	–	–	2.82	2.82	8.08	-1.04	4.78	1.040
42	168	GP	–	–	0.00	2.82	9.40	-1.04	4.78	1.040
43	169	GP	–	–	0.00	2.82	9.40	-1.04	4.78	1.040
44	170	GP	–	–	2.82	2.82	8.08	-1.04	4.78	1.040
45	182	GP	–	–	5.25	5.25	15.05	-2.14	8.16	1.040
46	183	GP	–	–	7.98	7.98	22.88	-3.25	12.40	1.040
47	189	GP	–	–	384.37	170.75	569.15	-46.67	209.45	1.040
48	196	GP	–	–	0.00	20.25	67.50	-7.49	34.36	1.040
49	197	GP	–	–	0.00	20.25	67.50	-7.49	34.36	1.040

7.4 Electric–Gas Coupling Elements

This subsection reports P2G units and compressors that couple the electric and gas subsystems.

Table 20: `P200_G135` P2G Units.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW
1	135	131	0.45	304.00	336.00
2	136	132	0.45	184.00	216.00
3	127	134	0.45	184.00	216.00
4	125	15	0.45	100.00	140.00
5	126	36	0.45	100.00	140.00
6	152	69	0.45	100.00	140.00
7	153	82	0.45	100.00	140.00
8	154	96	0.45	100.00	140.00
9	155	105	0.45	100.00	140.00

Table 21: `P200_G135` Compressor Units; E and G denote electric-driven and gas-driven compressors, respectively.

No.	In.	Out.	Type	Power bus	Ratio p.u.	Elec. power MW	No.	In.	Out.	Type	Power bus	Ratio p.u.	Elec. power MW
1	18	131	E	135	1.00–1.02	0–800	16	58	134	E	127	1.00–1.02	0–800
2	77	106	G	136	1.00–1.02	–	17	57	118	G	142	1.00–1.02	–
3	67	107	G	135	1.00–1.02	–	18	57	119	G	148	1.00–1.02	–
4	67	108	E	136	1.00–1.02	0–800	19	45	120	E	135	1.00–1.02	0–800
5	64	109	G	125	1.00–1.02	–	20	45	121	G	136	1.00–1.02	–
6	64	110	G	126	1.00–1.02	–	21	45	122	G	135	1.00–1.02	–
7	35	132	E	127	1.00–1.02	0–800	22	45	123	E	136	1.00–1.02	0–800
8	35	133	G	142	1.00–1.02	–	23	31	124	G	125	1.00–1.02	–
9	6	111	G	148	1.00–1.02	–	24	31	125	G	126	1.00–1.02	–
10	6	112	E	135	1.00–1.02	0–800	25	60	126	E	127	1.00–1.02	0–800
11	6	113	G	136	1.00–1.02	–	26	60	127	G	142	1.00–1.02	–
12	114	135	G	135	1.00–1.02	–	27	50	128	G	148	1.00–1.02	–
13	115	135	E	136	1.00–1.02	0–800	28	72	129	E	135	1.00–1.02	0–800
14	58	116	G	125	1.00–1.02	–	29	72	130	G	136	1.00–1.02	–
15	58	117	G	126	1.00–1.02	–							

8 Reproducibility Note

The benchmark descriptions in this companion document are intended to be read as a deterministic data specification for the three IEGS cases. The tabulated gas wells, gas-network nodes, ordinary pipelines, power-side generation units, P2G units, and compressors are the effective benchmark data used by the corresponding integrated electric–gas OPF and reinforcement-learning studies. The same notation, units, and coupling conventions are used across `P14_G11`, `P57_G40`, and `P200_G135`, so that differences among the three cases reflect benchmark scale and topology rather than changes in data interpretation.